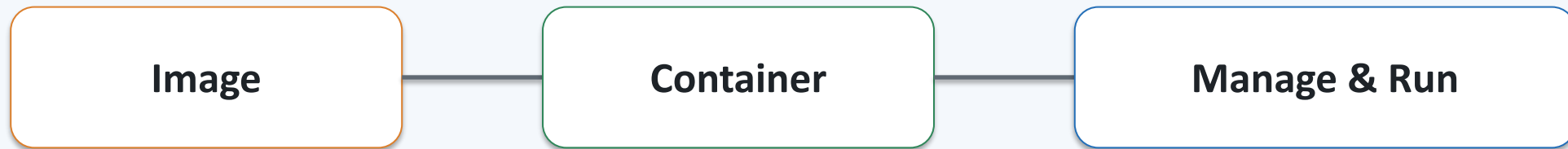


Docker Part 2: Essential Docker Commands



Learning Objectives

Use basic Docker commands to manage images and containers.

Explain the container lifecycle: create/run, stop, start, and remove.

Access a running container and inspect its information or logs.

Publish a container port to the host.

Recognize basic file-copy and bind-mount commands used in later practicals.

Check Docker Before Starting

Run these commands in the Ubuntu terminal.

```
docker --version  
docker ps  
docker images
```

```
sudo systemctl status docker
```

docker --version → check Docker installation/version

docker ps → show running containers

docker images → show local images

Use this if you need to check whether the Docker service is running.

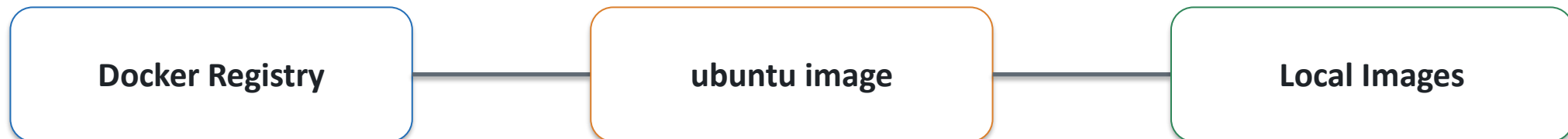
Image Management Commands

```
docker pull ubuntu  
docker images  
docker rmi ubuntu
```

docker pull IMAGE → download an image

docker images → list images stored locally

docker rmi IMAGE → remove an image that is no longer needed



Create and Run a Container

```
docker run --name u1 -it ubuntu bash
```

docker run → create and start a new container

--name u1 → give the container a readable name

-i → keep standard input open

-t → allocate a terminal

ubuntu → image name

bash → command/shell to run inside the container

Ubuntu Image

**u1 Container
Running bash**

View Containers

```
docker ps  
docker ps -a
```

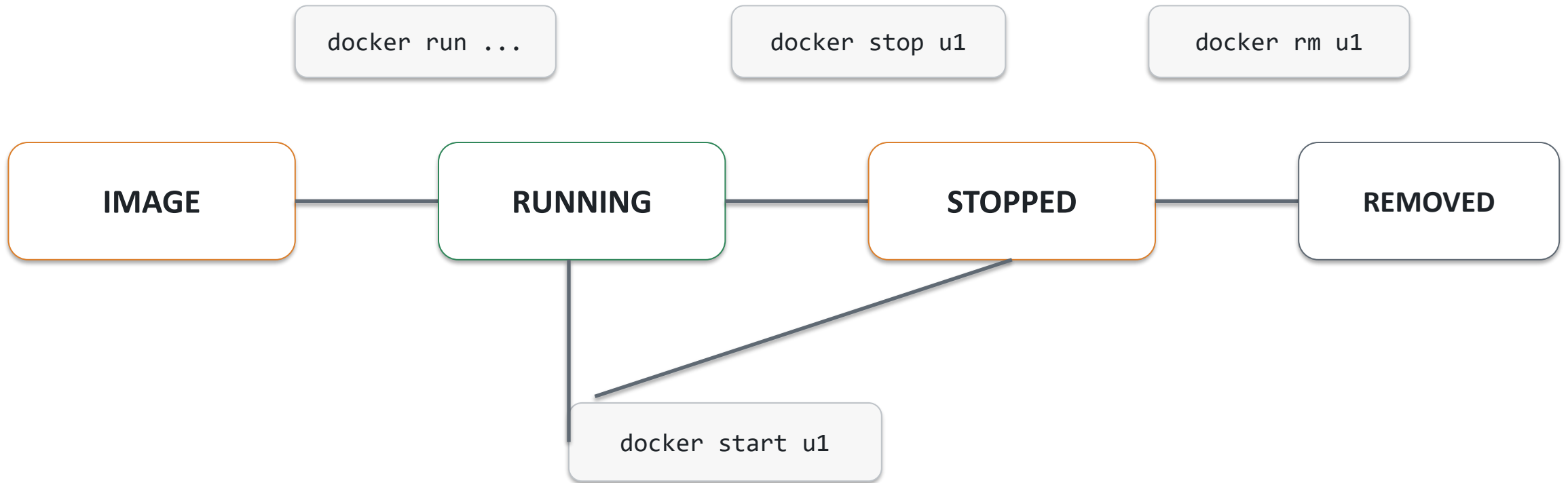
docker ps → running containers only

docker ps -a → all containers, including stopped/exited containers

Running
Visible in docker ps

Stopped / Exited
Visible in docker ps -a

Container Lifecycle



Important: `docker run` creates a new container; `docker start` restarts an existing stopped container.

Access a Running Container

```
docker start u1  
docker exec -it u1 bash
```

docker exec runs a new command inside an already running container.
docker exec -it u1 bash is a clear way to open an interactive shell.
Type exit to leave the shell.

```
docker attach u1
```

docker attach connects to the container's main process. For beginners...

Inspect and Troubleshoot

```
docker inspect u1  
docker logs u1
```

docker inspect → detailed configuration and runtime information

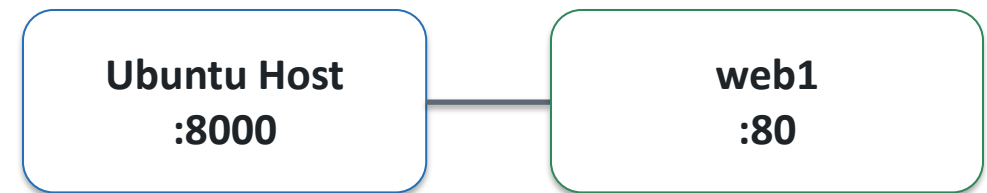
docker logs → output produced by the container's main process

These commands are useful when checking how a container is configured

Detached Mode and Port Mapping

```
docker run -d --name web1 -p 8000:80 httpd
```

- d → run in the background (detached mode)
- p 8000:80 → Host Port 8000 → Container Port 80
- name web1 → name the container
- httpd → Apache HTTP Server image



```
http://localhost:8000
```

Copy a File into a Container

```
docker cp index.html web1:/usr/local/apache2/htdocs/index.html
```

docker cp copies files or directories between the host and a container.

This is useful for a simple demonstration, such as replacing Apache's index.html.

For continuously shared web content, a bind mount is more convenient.



Bind Mount: Share Host Files with a Container

```
docker run -d --name web2 -p 9000:80 \
-v /webdata:/usr/local/apache2/htdocs \
httpd
```

-v HOST_PATH:CONTAINER_PATH → mount a host directory

Changes made to files in the host directory are visible through the container

Example browser URL: <http://localhost:9000>



Essential Commands: Quick Reference

```
docker pull IMAGE
```

Download image

```
docker images
```

List local images

```
docker run ...
```

Create + start container

```
docker ps
```

Running containers

```
docker ps -a
```

All containers

```
docker stop NAME
```

Stop container

```
docker start NAME
```

Start existing container

```
docker rm NAME
```

Remove container

```
docker exec -it NAME bash
```

Open shell in running container

```
docker logs NAME
```

View logs

```
docker inspect NAME
```

View detailed information

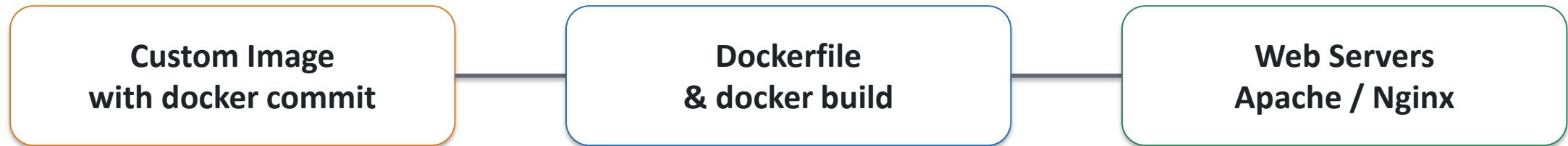
```
docker rmi IMAGE
```

Remove image

Suggested Hands-on Practice

1. Check Docker and list images/containers.
2. Pull the Ubuntu image.
3. Create an interactive Ubuntu container named u1.
4. Create a file inside u1, exit, then inspect container status.
5. Start u1 again and access it with docker exec.
6. Stop and remove u1.
7. In the next practical, deploy Apache/Nginx with port mapping and bind mounts.

Next Lesson



After understanding the essential commands, move to reproducible custom images with Dockerfile and practical web-server deployment.